22 Reactor Design for Multiple Reactions

Why Are We Learning This?

In earlier reactor-design problems, we usually studied a single reaction. While those problems helped us learn mole balances, reactor models, and kinetics, most real chemical reactors involve multiple reactions occurring simultaneously. Some reactions produce desired products, while others produce undesired byproducts, consume valuable reactants, generate hazardous compounds, or create thermal-safety concerns.

As chemical engineers, we must account for all reactions that can occur—not just the reaction we want. Even reactions that are normally suppressed may become important during startup, shutdown, upset conditions, or temperature excursions. Many industrial reactor failures have occurred because secondary reactions were ignored during reactor design.

By the end of this lesson, you should understand:

- The major types of multiple-reaction networks.
- The difference between reaction rates and species rates.
- Why conversion is usually not sufficient for multiple-reaction problems.
- How to write net species rates.
- Why mole balances must be written using species rates rather than reaction rates.
- How multiple reactions affect reactor design, selectivity, yield, and safety.

Course Roadmap

Introduction to CRE
↓
General Balance Equation
↓
Common Reactor Models
(CSTR, Batch, PFR)
↓
Reaction Progress

(Conversion)
↓
Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)
↓
Rate Laws
(Kinetics)
↓
Equilibrium
(Reversible Reactions)
↓
Equilibrium Conversion
(Thermodynamic Limitations)
↓
One Reaction: Solving in terms of Reaction Progress
↓
One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)
↓
Catalytic Reactions
↓
Numerical Methods
(Python)
↓
Energy Balances
↓
One Reaction: Non-Isothermal Reactor Design
↓
►Multiple Reactions◄
↓
Reactor Safety
↓
Challenging the Well-Mixed Assumption
(Residence Time Distributions)
↓
Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Big Idea

Single-reaction problems teach reactor design. Multiple-reaction problems teach reactor engineering. Most industrial reactors contain competing reactions, so understanding reaction networks is essential for designing safe and profitable processes.

Multiple reaction definitions:

- Parallel:

Rate laws:

- Series:

Rate laws:

- Independent:

Rate laws:

- Complex:

Rate laws:

- How does this change our procedure for solving reactor design problems?
    -

Need to include all relevant reactions and account for stoichiometry in the different reactions.

    -

For the most part, you have to use N or F. The only exceptions that I can think of are:

1. A single reversible reaction, e.g., $A \leftrightharpoons B$
2. A single series of reactions, e.g., A --> B --> C.

The two reactions listed above can be solved in terms of X. All other problems involving multiple reactions have to use N or F. The reason for this is:

Why Conversion Usually Does Not Work for Multiple Reactions

Conversion works well when there is only one reaction because all species can be related to a single variable.

For example:

A → B

Once X is known CA and CB can be calculated directly.

However, for multiple reactions:

A → B + C

C + A → D

the amount of C depends on both how fast C is produced and how fast C is consumed. As a result, a single conversion variable is usually insufficient.

Instead, we must track species individually using N or F (moles or molar flow rates).

Let's do an example: The following reactions are taking place in liquid phase in a CSTR:

3A --> B + C $\qquad -r_{1A} = k_{1A}C_A, k_{1A} = 7.0 \text{ min}^{-1}$

2C + A --> 3D $\qquad -r_{2D} = k_{2D}C_C^2 C_A, k_{2D} = 3.0 \frac{\text{L}^2}{\text{mol}^2\text{min}}$

4D + 3C --> 3E $\qquad r_{3E} = k_{3E}C_D C_C, k_{3E} = 2.0 \frac{\text{L}}{\text{mol min}}$

Write expressions for the net rates of A, B, C, D, and E, and use them to calculate the steady state rates in a CSTR that is operating at C_A = 0.10 M, C_B = 0.93 M, C_C = 0.51 M, C_D = 0.049 M.

Reaction Rates vs Species Rates

This is the most important idea in this set of notes.

Reaction Rates: describe how quickly reactions occur.

r1, r2, and r3 are reaction rates.

Species Rates: describe how quickly species are produced or consumed.

rA, rB, rC, rD, and rE are species rates.

These are usually not the same.

For example:

Reaction 1: $A \rightarrow B + C$

Reaction 2: $C + A \rightarrow D$

Species C is produced in Reaction 1 and consumed in Reaction 2. Therefore:

Net Rate of C = Formation Rate − Consumption Rate

Mole balances always use species rates not reaction rates.

Why Do We Need Net Species Rates?

A molecule does not "know" which reaction created it. The reactor only experiences the net effect of all reactions occurring simultaneously.

For example:

Species D: Produced in Reaction 2, Consumed in Reaction 3

The reactor therefore responds to: Net Rate of D = Production − Consumption

This is why reactor mole balances are written using net species rates rather than individual reaction rates.

Let's take this same example. If the volumetric flowrate is 100 L/min and the entering concentration of A is 3 M, what is the volume of the CSTR?

Summary and Key Takeaways

Today we introduced reactor design for systems involving multiple reactions.

Key Ideas

- ☑ Most industrial reactors involve multiple reactions.
- ☑ Desired and undesired reactions must both be considered.
- ☑ Multiple-reaction problems usually require moles or molar flow rates rather than conversion alone.
- ☑ A species may be produced in one reaction and consumed in another.
- ☑ Mole balances use net species rates, not reaction rates.
- ☑ Stoichiometry must be applied across the entire reaction network.

Most Important Idea

Reaction Rates
↓
Determine Species Rates
↓
Species Rates
↓
Appear in Mole Balances and
↓
Determine Reactor Performance

The central challenge of multiple-reaction reactor design is tracking how species are simultaneously produced and consumed by different reactions. Once the net species rates are known, reactor-design procedures follow the same principles used for single-reaction systems.